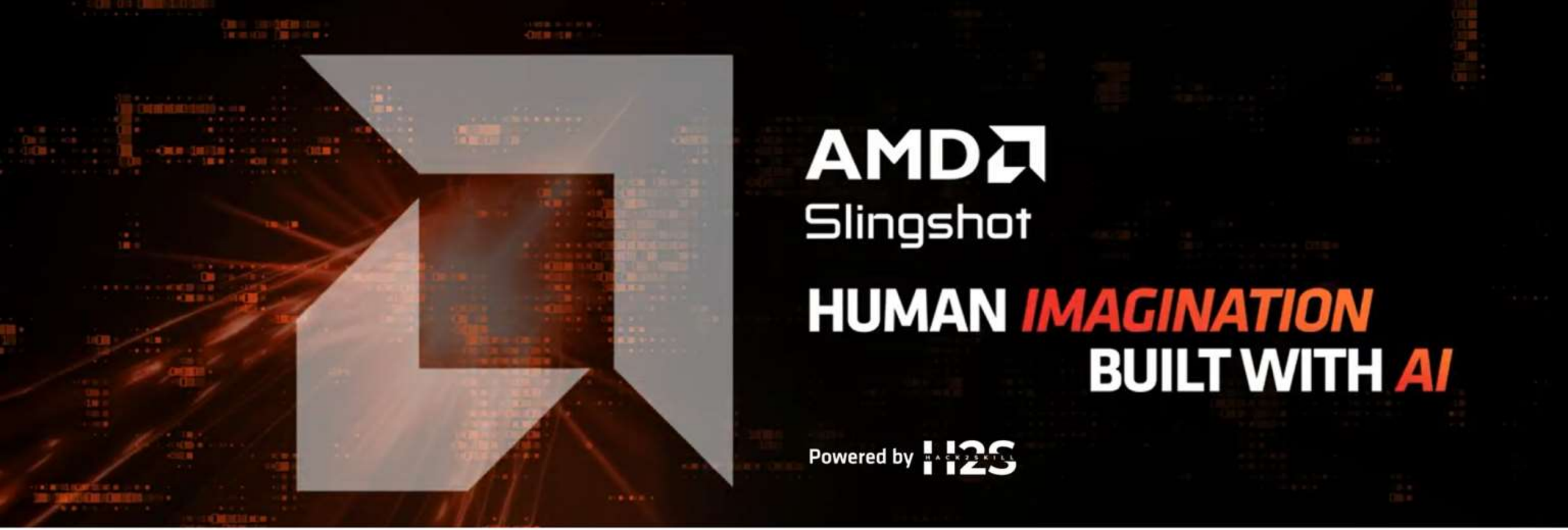
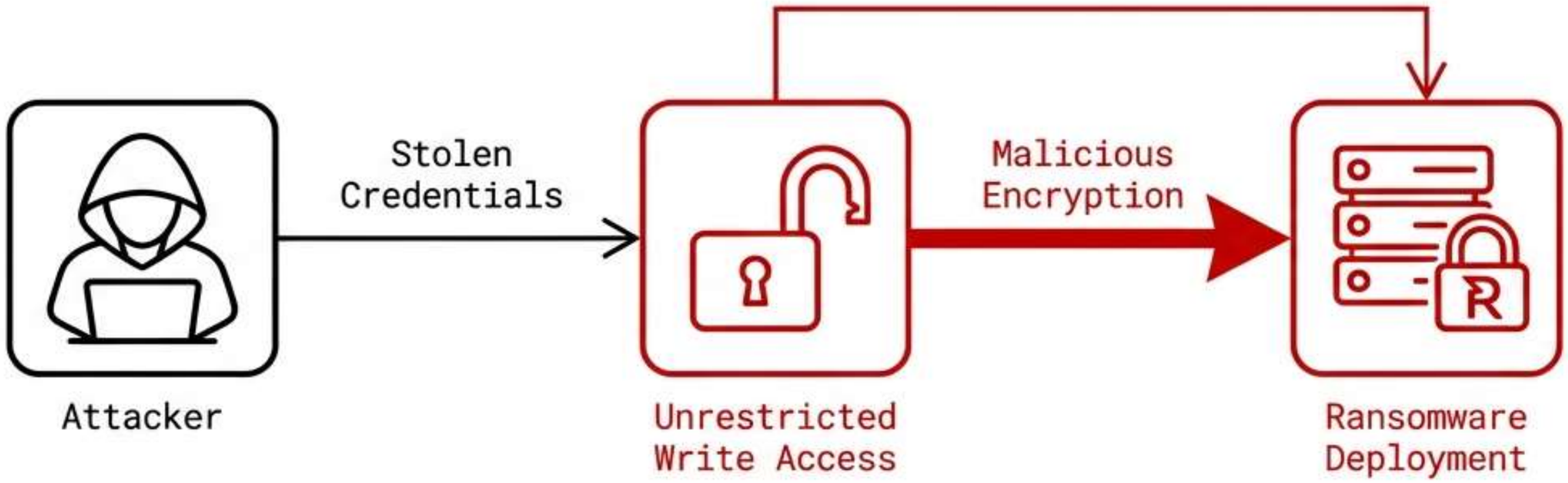


Team Details

- Team name – Eye
- Team leader name – Jay Choukikar
- Problem Statement - **AI + Cybersecurity & Privacy**

SentriKey: Hardware-Enforced Zero-Trust Write Architecture
Moving from passive vulnerability to active verification.



The Core Thesis:

Traditional systems grant write permissions passively.

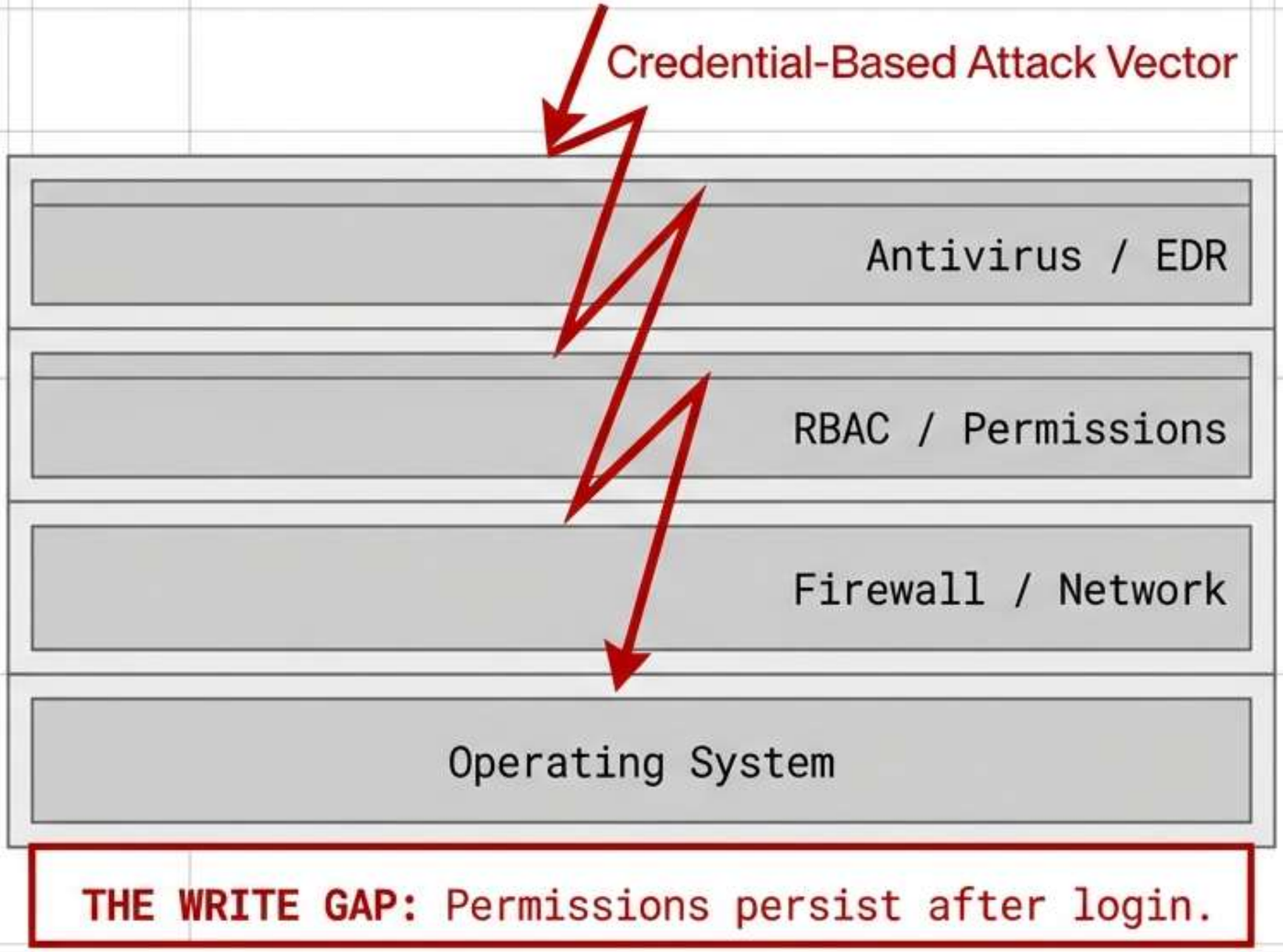
SentriKey transforms write access into a high-friction, verified event.

Ransomware is not just a malware problem; it is an access control problem.

Context:
Global ransomware costs are projected to exceed $265 billion annually by 2031

The Limits of Software-Only Defences

- **Credential Vulnerability:** Valid credentials allow attackers to bypass standard firewalls and Role-Based Access Control (RBAC). The system assumes the user is legitimate.
- **The Backup Problem:** Ransomware specifically targets network backups because write access remains persistent across the connected environment.
- **The Reactive Gap:** Antivirus and EDR (Endpoint Detection and Response) tools typically react *after* the breach or encryption process has already begun.



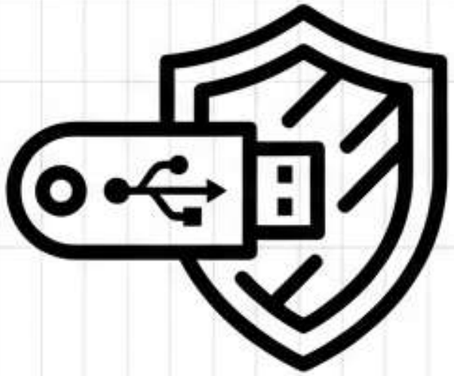
The SentriKey Shift: Active Verification

TRADITIONAL MODEL

State: **Always Open**
Permissions granted once at login.

Result: High attack surface. Once inside, the attacker has free reign.

SENTRIKEY MODEL



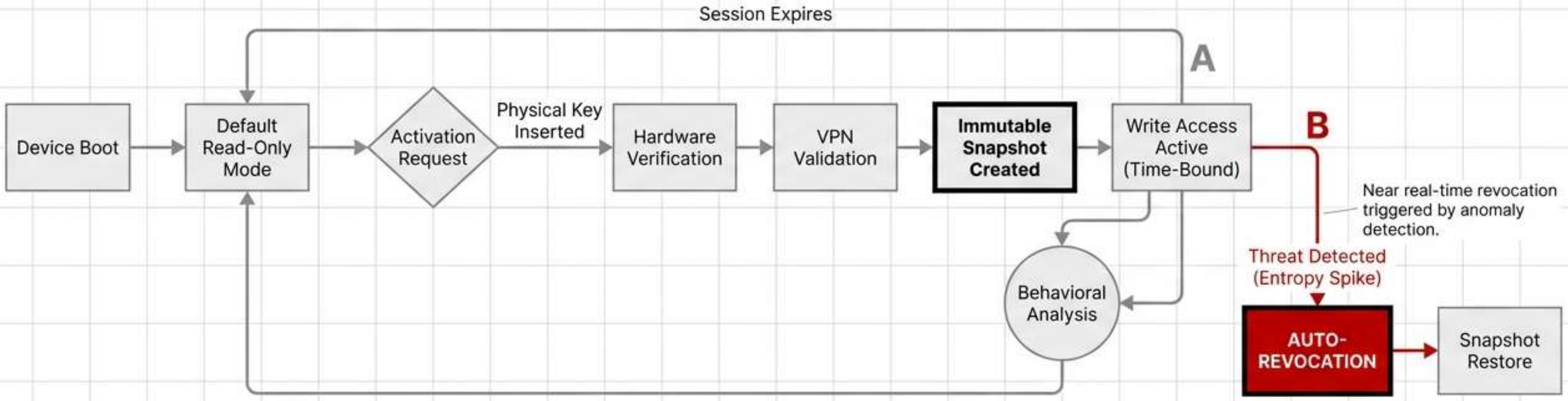
State: **Read-Only by Default**
Write access requires physical hardware validation.

Constraints: Time-bound sessions with multi-layer verification.

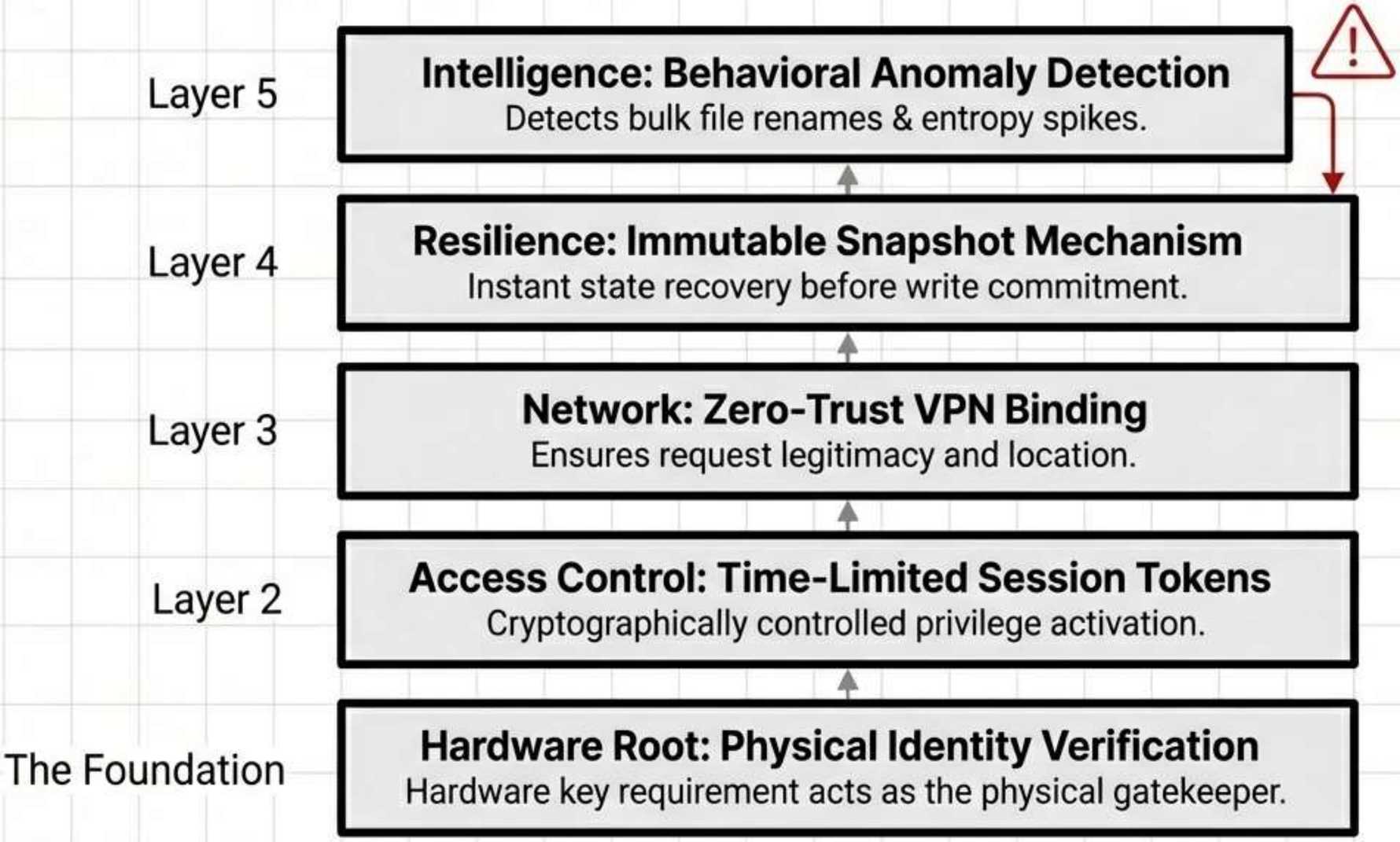
Result: Significantly reduced attack surface.

We do not rely on user trust; we rely on physical interaction and cryptographic verification.

The Lifecycle of a Secure Write Session

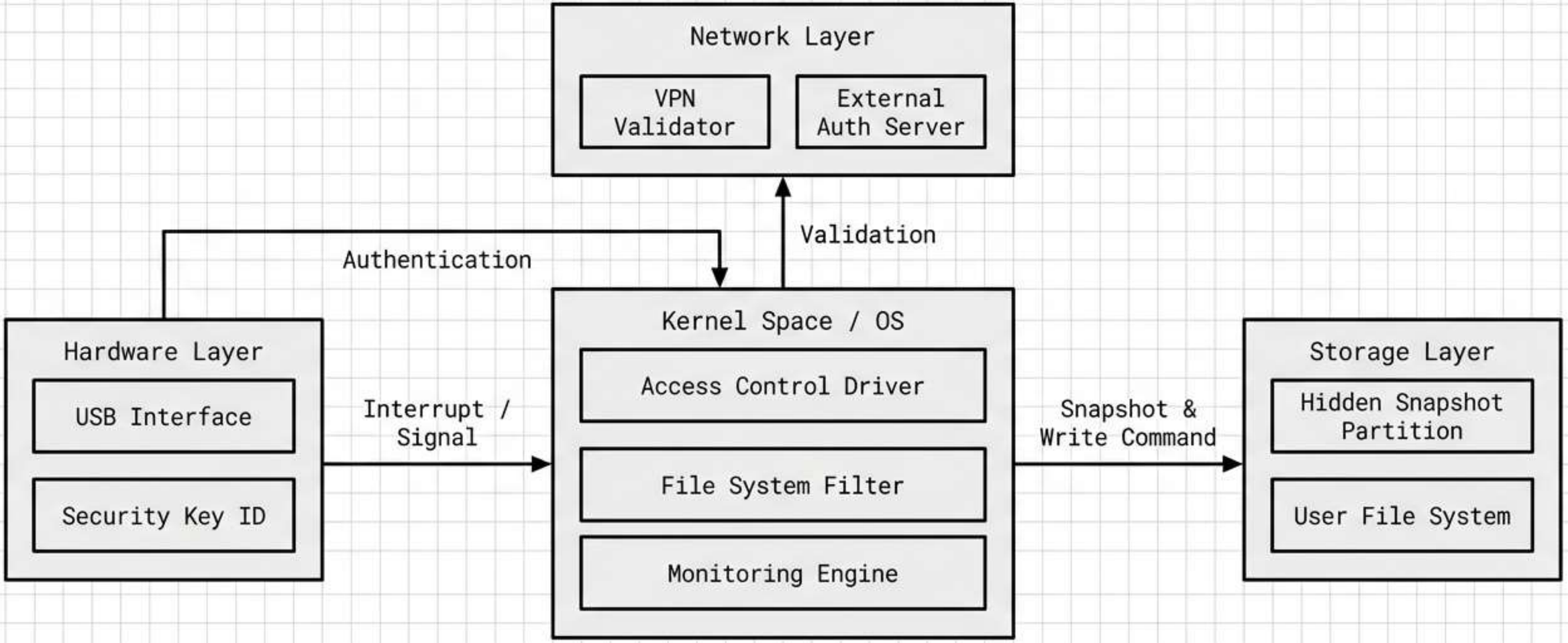


A Layered Defence Stack

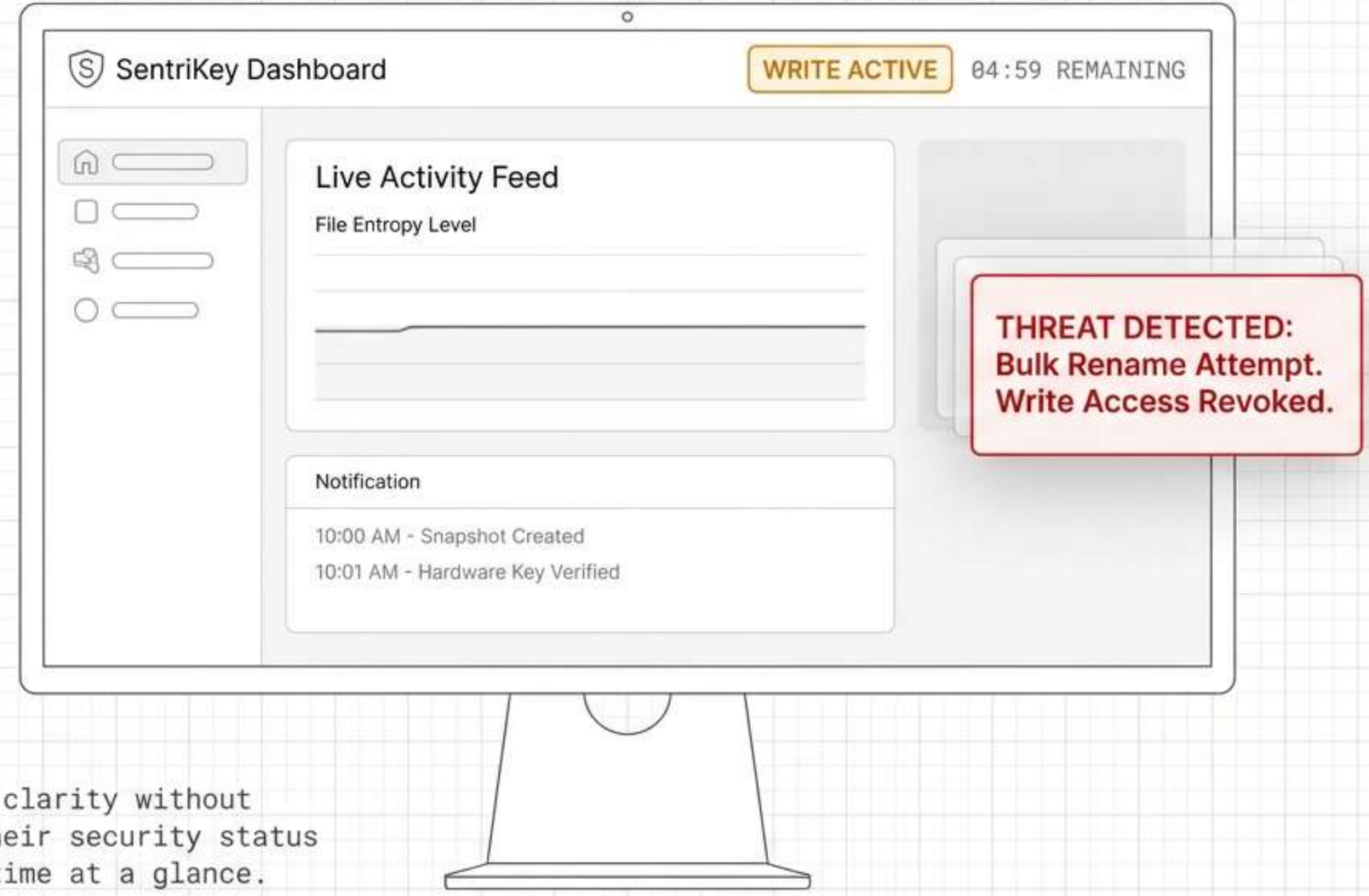


This architecture ensures that even if one layer is bypassed, the physical and cryptographic dependencies prevent a successful mass-encryption event.

System Architecture Schematic



User Experience & Admin Control



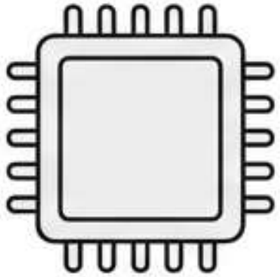
The interface provides clarity without intrusion. Users see their security status and remaining session time at a glance.

Differentiation & Added Value

Feature	Antivirus / EDR	RBAC / MFA	**SentriKey**
Hardware-Gated **Write Control**	○	○	✓
Time-Bound Privilege Sessions	○	◐	✓
Behavioral Write Revocation	✓ (Detection only)	○	✓ (Revocation)
Snapshot Containment	◐	○	✓

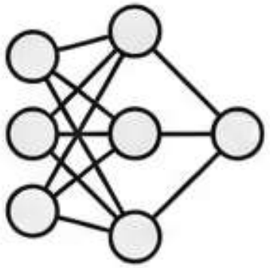
While EDR focuses on detection and RBAC focuses on identity, only SentriKey enforces hardware constraints on the write action itself, reducing the attack surface significantly.

Powering Scale with AMD Integration



AMD Ryzen™ / EPYC™ Processors

Utiilized for scalable beharal monitoring and efficient anomaly detection workloads. High core counts allow real-time entropy analysis without system lag.



AMD AI Acceleration Stack

Enables low-latency on-device inference. Allows the system to detect ransomware patterns locally, reducing reliance on cloud-based detection.

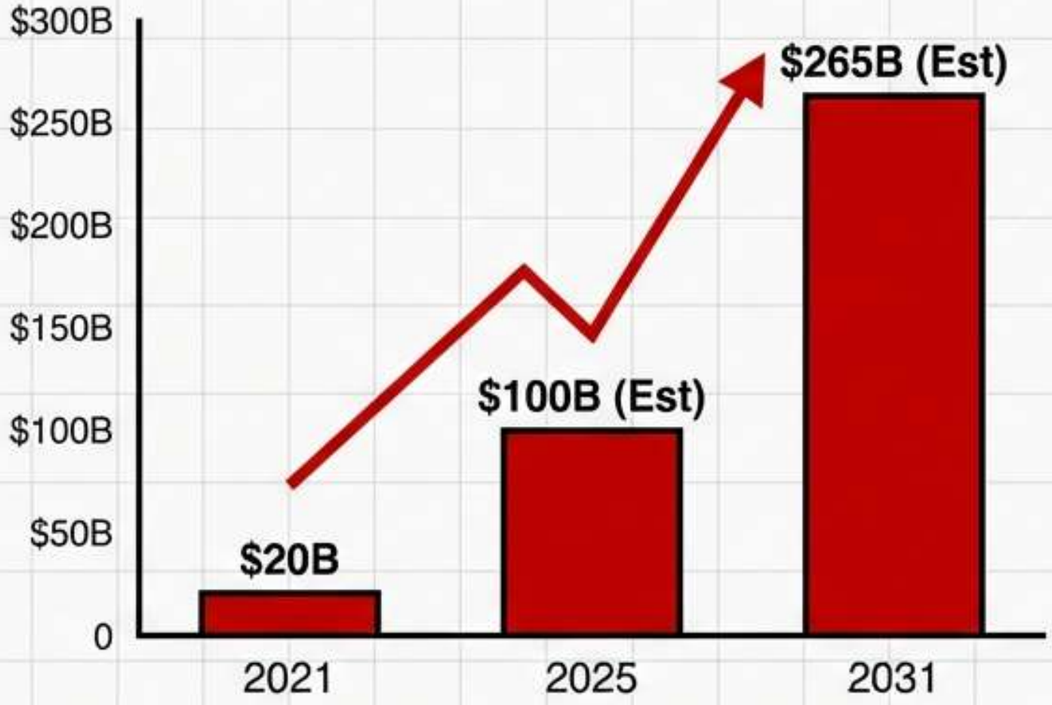
AMD Secure Processor (Future Roadmap)

Proposed integration to strengthen the hardware root-of-trust and support secure firmware attestation in enterprise deployments.

Current prototype demonstrates architecture principles. AMD integration represents secure production-scale deployment alignment.

Market Opportunity & Economic Imperative

Global Ransomware Damage Costs (USD Billions)



Target Sectors

Healthcare
Critical Patient Data

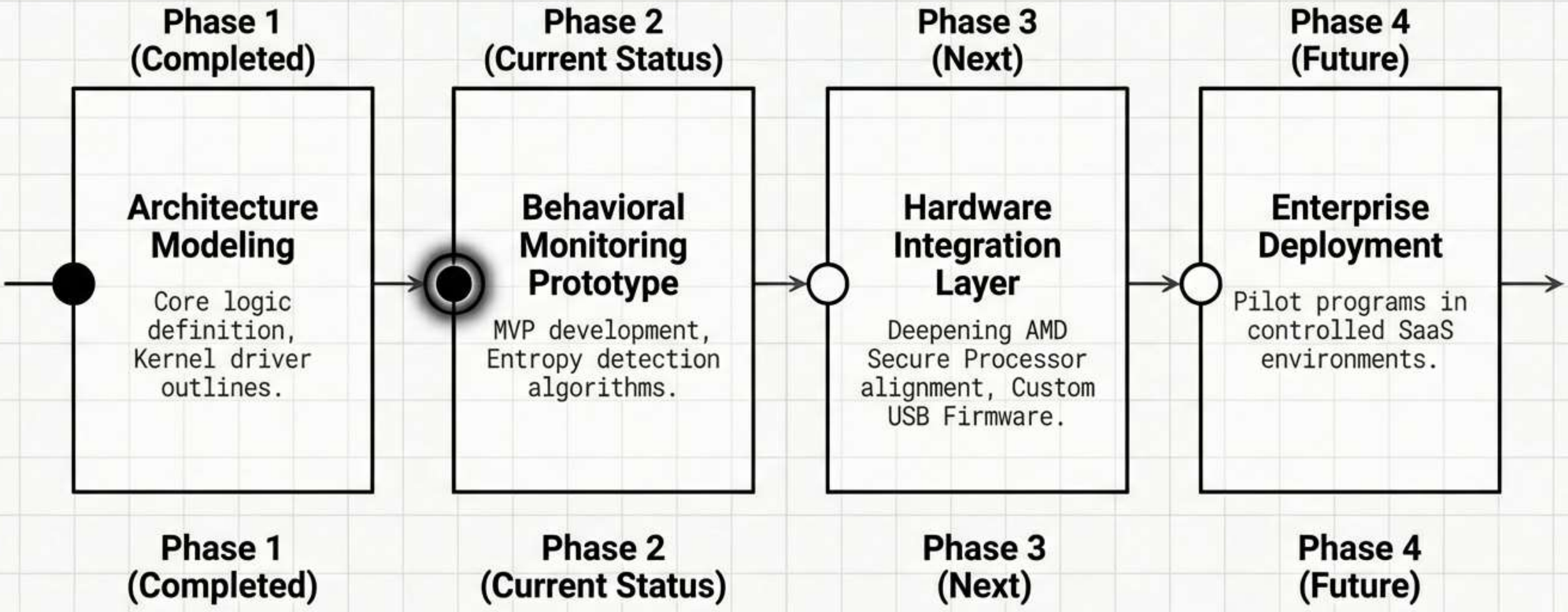
Finance
High-Frequency Transaction Integrity

Government
National Security Assets

SaaS/Cloud
Infrastructure Reliability

Companies are spending billions on recovery. SentriKey offers prevention at the architectural level. Scalable from single-device protection to enterprise network deployment.

Implementation Roadmap



Summary & Impact

- **The Flaw:** Persistent write access is the root **vulnerability** in modern computing.
- **The Fix:** Hardware-gated write control renders mass encryption economically infeasible.
- **The Result:** SentriKey introduces a model of active, monitored privilege.

Write Nothing. Trust Everything.

AMD Slingshot

HUMAN *IMAGINATION*
BUILT WITH *AI*

Powered by H2S HACKSKILL

Thank you!